

Original zeta, faithful source recovery, and mixed boundaries

Programme derivations with the cited human-source provenance

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The preceding theta-origin explanation required correction: lifting the coefficients of an imported classical theta kernel does not derive that kernel from the whole absolute base. Chapter 1 gives the original zeta and shifted-Hurwitz comparison, retaining its multiplier, both endpoints, embedded local modules, product, and support labels.

Chapter 2 computes the complete algebraic kernel of actual radial adelic periodization. Its prime-boundary coordinates and explicit inverse give a faithful joint receiver. This receiver is a map on the specified original source; it is not an asserted identification with the whole absolute geometry or its spectral quotient. The higher mixed-prime connecting classes and the two real-idele endpoint characters are proved with their original signs and factors.

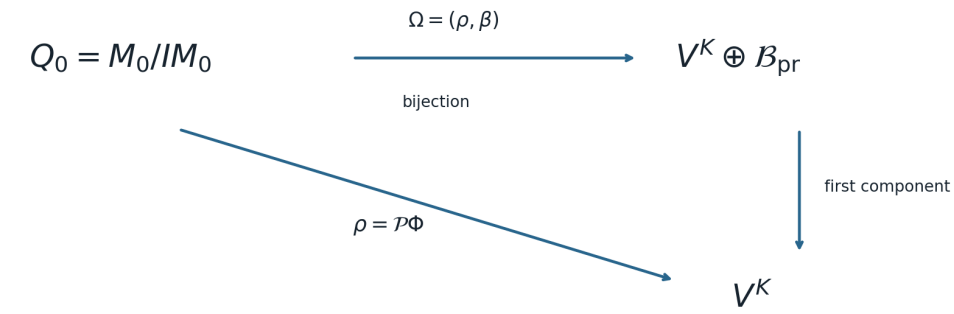
Chapter 3 gives the complete finite rational mixed-boundary criterion, including a sharp number of scalar equations for its obstruction space, an actual adelic antecedent, both endpoint values, compact-unit characters and the correction for separately labelled initial terms. No statement of positivity for every Weil test, or proof or disproof of RH, is made. The complete programme predecessor proof sources accompany these new derivations. Human primary sources are linked at their original hosts.

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The source is recovered by retaining its prime boundary

Exact algebraic maps; arbitrary nontrivial bounded distributive lattice L .



$$\ker \rho = B_{\text{pr}} = \bigoplus_{p \text{ prime}} \mathbb{C}^2$$

$$\beta([\sum_a t_a \otimes h_a]) = \sum_p \ell_p \otimes \sum_a v_p(a) (h_a(0), \int_{\mathbb{R}} h_a(x) dx)$$

$$\Omega^{-1}(F, b) = [t_1 \otimes P^{-1}F] + j(b)$$

$$ph(y) = 2 \sum_{n \geq 1} h(ny), \quad \mathcal{M}(Ph)(s) = 2\zeta(s)\mathcal{M}_+ h(s) \quad (\Re s > 1)$$

$$G_L(\rho)(c, 1_L) = e \neq \tau \quad (0 \neq c \in \ker \rho)$$

Proof: ABR1-19, ABR29-30. Sources: Connes, math/9811068v1, III(6)-(19); Connes, Consani and Marcolli, math/0703392v1, actual restriction image. One common label per object.

Figure 1: The joint map retains all algebraic source information. Periodization alone loses exactly the displayed prime boundary. The lifted zero image retains its support label.

A finite exact test of every mixed boundary coefficient

$$S(y) = \sum_{k \geq 1} v_k g(ky/B) + \sum_{k=1}^K d_k g(ky/B)$$

v is Q -periodic; d retains every omitted initial frequency; B retains the original scale.

$$M = K + \lfloor (Q - 1)/2 \rfloor$$

$$C_1 = \dots = C_M = 0 \iff C_m = 0 \ (m \geq 1)$$

$$\iff d_k = 0 \ (1 \leq k \leq K), \quad v_r = v_{Q-r} \ (1 \leq r < Q)$$

The two endpoint values survive the complete boundary cancellation:

$$\int_0^\infty S(y) \frac{dy}{y} = \frac{v_0}{4}, \quad \int_0^\infty S(y) dy = \frac{B}{4Q} \sum_{r=1}^Q v_r$$

$$\tilde{v}_k + \tilde{d}_k = \tilde{w}_k + (0, \sigma_k)$$

Separate input labels retain the omitted-initial support even when its amplitude cancels.

Proof: RMK1-15, RMK17-26, RMK30-32; original full Gaussian polynomial g is defined in RMK1.
This is a complete finite-rational boundary classification, not an all-tests positivity theorem.

Figure 2: A finite test determines the entire boundary in the stated rational family. Its zero-boundary functions retain two distinct endpoint values and their original support contributions.

1 Original zeta and the retained theta comparison

Correction and provenance. The statement that the Rodgers–Tao heat kernel was the programme’s original construction over the entire absolute base was not established. THS1–31 begin with that classical analytic kernel and construct a support-preserving coefficient lift. HTB1–35 establish its exact comparison with the shifted Hurwitz family. Neither construction is a derivation of the scalar kernel from the entire arithmetic geometry. The distinction is measurable: THS22 already proves that all supported zero times act identically on the actual theta orbit. The ambient coefficient injection does not make that orbit injective.

The original arithmetic functions in this calculation remain

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}, \quad F_t(s) = \sum_{n=0}^{\infty} (n+1+t)^{-s}, \quad \operatorname{Re} s > 1, \quad t > -1. \quad (\text{ZTC1})$$

Their continuations, endpoints, and exact shift generator are proved in HTB2–5, with the programme source identified there by version, hash, and original theorem labels. In particular $F_0 = \zeta$, $F_1 = \zeta - 1$, and $\partial_t F_t(s) = -sF_t(s+1)$. The latter family comes from the supplied April 2026 shifted-Hurwitz programme draft, not from a proposed identification of two time variables. Brad Rodgers and Terence Tao’s original author source, equations `phidef`, `htdef`, `hoz`, and `sas`, supplies the auxiliary heat kernel used in THS and HTB. These are cited inputs, not new discoveries. The following comparisons are derived explicitly. Their complete HTB, THS, and CFD proofs accompany this chapter.

1.1 A direct theta sum for the original arithmetic family

For real $t > -1$ and $x > 0$, retain the actual shifted frequencies and put

$$K_t(x) = \sum_{n=0}^{\infty} e^{-\pi(n+1+t)^2 x}. \quad (\text{ZTC2})$$

This is a scalar analytic representation of the stated arithmetic family. No assertion that it determines the full support lattice is part of this definition. For fixed $a = 1 + t > 0$, monotonicity of the summand as a function of n gives

$$K_t(x) \leq e^{-\pi a^2 x} + \int_0^{\infty} e^{-\pi(a+y)^2 x} dy \leq 1 + \frac{1}{2\sqrt{x}}. \quad (\text{ZTC3})$$

For $x \geq 1$ the same bound after retaining the first exponential gives $K_t(x) \leq C_a e^{-\pi a^2 x}$, since $(a+n)^2 - a^2 \geq n^2$ and $C_a = \sum_{n \geq 0} e^{-\pi((a+n)^2 - a^2)} < \infty$. Consequently the following integral converges absolutely for $\operatorname{Re} s > 1$:

$$\int_0^{\infty} x^{s/2-1} K_t(x) dx = \pi^{-s/2} \Gamma(s/2) F_t(s), \quad F_t(s) = \frac{\pi^{s/2}}{\Gamma(s/2)} \int_0^{\infty} x^{s/2-1} K_t(x) dx. \quad (\text{ZTC4})$$

Indeed, with $\sigma = \operatorname{Re} s > 1$, the sum of the absolute termwise integrals equals $\pi^{-\sigma/2} \Gamma(\sigma/2) \sum_{n \geq 0} (n+a)^{-\sigma}$. Fubini therefore applies. Substitution $y = \pi(n+a)^2 x$ in each term proves the displayed identity with its full Gamma and pi factors. Continuation outside that domain means the meromorphic continuation proved in HTB4, not an assertion that the unsubtracted integral converges there.

There is also an exact inverse recovering the kernel, not only the arithmetic function. For every real $c > 1$,

$$K_t(x) = \frac{1}{4\pi i} \int_{c-i\infty}^{c+i\infty} \pi^{-s/2} \Gamma(s/2) F_t(s) x^{-s/2} ds. \quad (\text{ZTC4a})$$

Here the integral is absolutely convergent. To prove the inversion with its constant, set $\sigma = c/2 > 0$ and $g(v) = \exp(\sigma v - e^v)$. This function and all its derivatives are integrable on $\mathbb{R}$: at $-\infty$

they are bounded by a constant times $e^{\sigma v}$, and at $+\infty$ by a polynomial in e^v times $g(v)$. The substitution $u = e^v$ identifies its Fourier transform with $\Gamma(\sigma - iy)$. Repeated integration by parts therefore bounds that Gamma factor by $C_N(1 + |y|)^{-N}$ for every fixed N . Fourier inversion gives $e^{-u} = (2\pi i)^{-1} \int_{\sigma - i\infty}^{\sigma + i\infty} \Gamma(w) u^{-w} dw$. One may verify this inversion directly by first multiplying the Fourier integrand by $e^{-\epsilon y^2}$: the resulting expression is the convolution of g with the Gaussian approximate identity, which converges to g ; the stated integrable transform bound permits dominated convergence as $\epsilon \downarrow 0$. Apply the identity to $u = \pi(n + 1 + t)^2 x$, then put $s = 2w$. The resulting factor is $1/(4\pi i)$. Finally sum in n . Interchange is justified by $\sum (n + 1 + t)^{-c} < \infty$ and the integrable Gamma bound. This proves ZTC4a. ZTC4 and ZTC4a are mutual recovery formulas for the specified original arithmetic family and its scalar theta sums; they do not recover a support lattice from a scalar function.

Differentiation in t is locally uniform for t in compact subsets of $(-1, \infty)$ and x in compact subsets of $(0, \infty)$, by Gaussian majorants for a polynomial times e^{-cn^2} . It gives the exact formula

$$\partial_t K_t(x) = -2\pi x \sum_{n=0}^{\infty} (n + 1 + t) e^{-\pi(n+1+t)^2 x}. \quad (\text{ZTC5})$$

For $\text{Re } s > 0$, integration of its absolute value is bounded by a constant times $\sum (n + a)^{-\sigma-1}$, locally uniformly in t . More precisely its Mellin transform is

$$\int_0^{\infty} x^{s/2-1} \partial_t K_t(x) dx = -2\pi^{-s/2} \Gamma(s/2 + 1) F_t(s + 1) = -s\pi^{-s/2} \Gamma(s/2) F_t(s + 1). \quad (\text{ZTC6})$$

Thus the original arithmetic shift generator is represented directly, without substituting the heat generator for it. The stated equality extends meromorphically by HTB4–5.

The separated sheet and its entire missing summand are visible before any transform:

$$K_1(x) = K_0(x) - e^{-\pi x}, \quad D_t(s) = \zeta(s) - F_t(s), \quad D_1(s) = 1, \quad (\text{ZTC7})$$

$$\pi^{-s/2} \Gamma(s/2) D_t(s) = \int_0^{\infty} x^{s/2-1} (K_0(x) - K_t(x)) dx \quad (\text{Re } s > 1). \quad (\text{ZTC8})$$

ZTC7 follows by deleting the $n = 1$ summand of K_0 ; ZTC8 follows by subtracting two absolutely convergent integrals. At $t = 1$ the integral is exactly $\pi^{-s/2} \Gamma(s/2)$, so its inverse returns the actual constant 1, including its amplitude and support when lifted.

1.2 Original zeta with both Mellin endpoints retained

At $t = 0$, HTB6 proves

$$K_0(x) = x^{-1/2} K_0(1/x) + \frac{1}{2} x^{-1/2} - \frac{1}{2}. \quad (\text{ZTC9})$$

Split ZTC4 at $x = 1$, apply this identity only to the integral over $(0, 1)$, and substitute $y = 1/x$ there. Initially $\text{Re } s > 1$, and direct integration gives

$$\zeta(s) = \frac{\pi^{s/2}}{\Gamma(s/2)} \left\{ \frac{1}{s-1} - \frac{1}{s} + \int_1^{\infty} K_0(x) \left(x^{s/2-1} + x^{-(s+1)/2} \right) dx \right\}. \quad (\text{ZTC10})$$

The two separate rational terms are the transforms of the two separate terms $x^{-1/2}/2$ and $-1/2$. The integral on $[1, \infty)$ is entire in s , since exponential decay bounds every logarithmic derivative on compact subsets of the s -plane. This proves the meromorphic continuation of the identity and keeps both endpoint contributions. Neither has been declared zero or absorbed into an unrecorded constant.

For the original cosine heat kernel, retain

$$\begin{aligned} A(s) &= \frac{s(s-1)}{2} \pi^{-s/2} \Gamma(s/2), & s(z) &= \frac{1}{2} + \frac{iz}{2}, & z(s) &= -2i(s - 1/2), \\ H_0(z) &= \frac{s(z)(s(z) - 1)}{16} \pi^{-s(z)/2} \Gamma(s(z)/2) \zeta(s(z)), \\ \zeta(s) &= \frac{16\pi^{s/2}}{s(s-1)\Gamma(s/2)} H_0(z(s)). \end{aligned} \quad (\text{ZTC11})$$

The proof is HTB6–10: the exact Mellin substitution and two integrations by parts yield the displayed multiplier, with all endpoint limits justified in the original convergence half-plane. The last equality is an identity of meromorphic functions. It is not a licence to replace the original zeta calculation by an entire function calculation while dropping its multiplier.

1.3 The boundary is part of the receiving object

Let $\mathcal{M}_s$ and $\mathcal{M}_z$ be the complex vector spaces of meromorphic functions on the respective complex planes. Retain the linear isomorphism, already proved in HTB11,

$$Uf(z) = \frac{A(s(z))}{8}f(s(z)), \quad U^{-1}G(s) = \frac{8G(z(s))}{A(s)}. \quad (\text{ZTC12})$$

Define the joint map on the entire two-component space by

$$\mathcal{V} : \mathcal{M}_s \oplus \mathcal{M}_s \longrightarrow \mathcal{M}_z \oplus \mathcal{M}_z, \quad (F, D) \longmapsto (U(F + D), UD). \quad (\text{ZTC13})$$

It has the exact inverse

$$\mathcal{V}^{-1}(H, J) = \left(\frac{8(H - J) \circ z}{A}, \frac{8J \circ z}{A} \right). \quad (\text{ZTC14})$$

Substitution proves both compositions equal the identity; all entries are meromorphic functions, so no undefined product of separate singular point values is involved. On the actual arithmetic family,

$$\begin{aligned} J_t &= UD_t, & \mathcal{V}(F_t, D_t) &= (H_0, J_t), \\ F_t(s) &= \frac{16\pi^{s/2}}{s(s-1)\Gamma(s/2)}(H_0(z(s)) - J_t(z(s))), \\ J_1(z) &= \frac{A(s(z))}{8}, & J_t(i) &= -\frac{t}{8}. \end{aligned} \quad (\text{ZTC15})$$

For the last equality use HTB5: $D_t(0) = \zeta(0) - F_t(0) = t$, $A(0) = -1$, and $z(0) = i$. In particular the retained pair determines the real parameter t by $t = -8J_t(i)$. All t give the same first component H_0 . Discarding J_t therefore identifies different actual arithmetic inputs. This is an explicit failure of injectivity on the actual programme family, not merely an example in an unrelated vector space. On the unrestricted space the summation map $(G, J) \mapsto G + J$ has exactly the kernel $\{(B, -B) : B \in \mathcal{M}_z\}$.

For the complete support carrier, use the common label on the direct sum:

$$G_L(\mathcal{M}_s \oplus \mathcal{M}_s) \xrightarrow{\mathcal{V}^\dagger} G_L(\mathcal{M}_z \oplus \mathcal{M}_z), \quad ((F, D), \lambda) \longmapsto (\mathcal{V}(F, D), \lambda). \quad (\text{ZTC16})$$

Here $G_L(V) = \{(0, \lambda) : \lambda \in L\} \cup (V \times \{1_L\})$ with its original joins and meets. Additivity, compatibility with the $G_L(\mathbb{C})$ action, carrier membership, and the inverse follow componentwise from linearity and ZTC14, as explicitly proved in CFD11–12. This is a bijection for every bounded distributive L . It does not silently replace the common label by independent labels on two product factors. In particular $e = (0, 1_L)$ and $\tau = (0, 0_L)$ remain distinct when L is nontrivial.

The exact multiplication comparison. The preceding maps were declared as linear and semimodule maps. Their injectivity does not identify the original multiplication with ordinary multiplication of their images. This can be checked exactly, without discarding the completion multiplier. Set

$$B(z) = \frac{A(s(z))}{8} = \frac{s(z)(s(z) - 1)}{16} \pi^{-s(z)/2} \Gamma(s(z)/2). \quad (\text{ZTC16a})$$

For meromorphic f, g on the s -plane the two products are

$$\begin{aligned} U(fg)(z) &= B(z)f(s(z))g(s(z)), \\ (Uf)(z)(Ug)(z) &= B(z)^2 f(s(z))g(s(z)). \end{aligned} \quad (\text{ZTC16b})$$

In particular $U1 = B$, not the ordinary constant unit. The factor B is nonconstant: it has a zero at $z(1) = -i$ and poles at $z(-2m)$, $m \geq 1$. Thus ordinary multiplication is not preserved. The exact transported product and its unit are

$$G \odot H = \frac{8G(z)H(z)}{A(s(z))}, \quad 1_{\odot}(z) = \frac{A(s(z))}{8}. \quad (\text{ZTC16c})$$

This formula is an identity of meromorphic functions, including their germs at the zeros and poles of the multiplier. It is not pointwise division of undefined singular values. Substitution gives $(Uf) \odot (Ug) = U(fg)$. Directly, $(G \odot H) \odot K = GHK/B^2 = G \odot (H \odot K)$, $B \odot G = G$, and distributivity follows from distributivity of meromorphic multiplication. Together with ZTC12 this proves a unital algebra isomorphism with the stated product.

For each local holomorphic ring $\mathcal{O}_{s_0}$ the affine coordinate map identifies it with $\mathcal{O}_{z(s_0)}$. Its exact image under U is the embedded fractional lattice $B\mathcal{O}_{z(s_0)}$. This lattice is closed under $\odot$, since $(Bu) \odot (Bv) = Buv$, and has unit B . Keeping instead the unmodified target $\mathcal{O}_{z(s_0)}$ can fail closure: at $s_0 = 1$, B has a simple zero, so $1 \odot 1 = 1/B$ is not holomorphic. This image-lattice statement differs from the preimage lattice $A^{-1}\mathcal{O}$ in ZTC17: the former transports an original holomorphic ring, whereas the latter specifies which original meromorphic sections have a holomorphic completed image.

The full support carrier admits the same exact comparison. Retain its original addition, and define the target product by

$$(G, \lambda) \odot_L (H, \mu) = \left(\frac{8GH}{A \circ s}, \lambda \wedge \mu \right), \quad 1_{\odot_L} = (B, 1_L). \quad (\text{ZTC16d})$$

If the amplitude of a product is nonzero, both input amplitudes are nonzero, hence both labels are 1_L ; this proves carrier closure. If either amplitude is zero its output amplitude is zero and the displayed meet retains its original label. Associativity and distributivity follow from ZTC16c and the meet/join laws of the bounded distributive lattice. The map $(f, \lambda) \mapsto (Uf, \lambda)$ and its inverse therefore give a unital semiring isomorphism from the original coefficient semiring to this specified target, preserving e and τ . An ideal P is sent to its literal image $U^\uparrow(P)$; closure under addition and multiplication follows by applying the isomorphism. Its inverse proves the converse. Moreover a product belongs to that image precisely when the corresponding original product belongs to P , proving that prime ideals correspond. This proves the exact spectrum comparison for these specified coefficient semirings. It is not a derivation of the entire arithmetic base from the scalar function H_0 .

For ZTC13–16, no multiplication on pairs was specified: the proved assertion there remains a common-label semimodule isomorphism. Neither ordinary multiplication nor a Weil pairing is silently inferred from that assertion.

1.4 Local zeros, poles, and the original trace

The earlier conversational assertion that the singular module map had not been established was also too broad: HTB13 already proves $A^{-1}\mathcal{O} \xrightarrow{U} \mathcal{O}$ with the affine coordinate change. CFD1–10 now make the comparison with the *original* regular lattice and every finite jet explicit. At a finite point let $h = s - s_0$ and $A = h^k a(h)$, $a(0) \neq 0$. Retain $I = A^{-1}\mathcal{O} = h^{-k}\mathcal{O}$ as an embedded submodule of the meromorphic germs, and $J = I \cap \mathcal{O}$. For nonzero $f \in I$ the original order is

$$\text{ord}_{s_0} f = \dim_{\mathbb{C}}(I/\mathcal{O}f) - \dim_{\mathbb{C}}(I/J) + \dim_{\mathbb{C}}(\mathcal{O}/J). \quad (\text{ZTC17})$$

This is proved by the power-of- h bases in CFD6–7, without identifying the reference lattices. At a trivial zero $s_0 = -2m$, $I = h\mathcal{O}$, $I/\mathcal{O}\zeta = 0$ and $\mathcal{O}/I$ has dimension 1. At $s_0 = 1$, $I = h^{-1}\mathcal{O}$, $I/\mathcal{O}\zeta = 0$ and $I/\mathcal{O}$ has dimension 1. Their signs in ZTC17 give, respectively, $+1$ and -1 . At $s_0 = 0$, $A(0) = -1$ is a unit. CFD4 provides the invertible triangular matrices recovering all Laurent coefficients with the exact factors 8 and $i/2$. Keeping only the zero algebra of the completed section

would give zero in both cancelled cases; keeping these comparison modules restores the actual divisor.

The earlier conversational explanation of the fibres requires a further correction. The map

$$\mathcal{O}/h\mathcal{O} \longrightarrow h\mathcal{O}/h^2\mathcal{O}, \quad [u] \longmapsto [hu] \quad (\text{ZTC17a})$$

is an isomorphism of $\mathcal{O}$ -modules. It is well-defined because $h(h\mathcal{O}) = h^2\mathcal{O}$, surjective by the definition of $h\mathcal{O}$, and injective because $hu \in h^2\mathcal{O}$ implies $u \in h\mathcal{O}$ in the local integral domain. Its inverse is $[hu] \mapsto [u]$. Consequently the two abstract fibres alone do not account for the loss. What must be retained is their specified embedding and the original reference lattice: multiplication by $A = h^{-1}a(h)$ identifies $I = h\mathcal{O}$ with $\mathcal{O}$, but the length-one quotient $\mathcal{O}/I$ records the cancelled original zero. Replacing the entire marked comparison by the completed section's zero quotient discards precisely that datum. Likewise multiplication by nonzero meromorphic A is an automorphism of the meromorphic sheaf as a module. It is not a ring automorphism for ordinary multiplication, as ZTC16b proves. It is not an automorphism of the fixed holomorphic lattice at its zeros and poles. These category statements do not contradict one another.

Logarithmic differentiation gives the original trace identity

$$\frac{\zeta'(s)}{\zeta(s)} = -2i \frac{H'_0(z(s))}{H_0(z(s))} - \frac{1}{s} - \frac{1}{s-1} + \frac{1}{2} \log \pi - \frac{1}{2} \psi(s/2). \quad (\text{ZTC18})$$

It first holds away from zeros and poles and then meromorphically; $\psi = \Gamma'/\Gamma$. For a positively oriented finite contour γ avoiding all zeros and poles of the functions and all singularities of every displayed integrand, and a holomorphic test g on and inside it, multiplication by g and integration gives

$$\begin{aligned} \frac{1}{2\pi i} \int_{\gamma} g \frac{\zeta'}{\zeta} ds &= \frac{1}{2\pi i} \int_{\gamma} g(s) (-2i) \frac{H'_0(z(s))}{H_0(z(s))} ds \\ &\quad - \frac{1}{2\pi i} \int_{\gamma} g(s) \left(\frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2) \right) ds. \end{aligned} \quad (\text{ZTC19})$$

At every zero or pole, write a meromorphic function as $(s - s_0)^r$ times a holomorphic unit. Its logarithmic derivative then has residue r . The residue theorem proves that ZTC19 counts the original zeros with multiplicity and the original poles with negative multiplicity, including every trivial zero inside the contour. No infinite-contour limit or decay assumption is hidden here. Extending this finite identity to a particular infinite trace requires the actual convergence estimates of that trace.

1.5 Two different trajectories through the same original zeta

The original- s function associated with the classical heat path is

$$f_h^{\text{heat}}(s) = \frac{16\pi^{s/2}}{s(s-1)\Gamma(s/2)} H_h(z(s)), \quad f_0^{\text{heat}} = \zeta. \quad (\text{ZTC20})$$

CFD18–21 give its full receiving operator and prove

$$\partial_h f_h^{\text{heat}}(-2) \Big|_{h=0} = 0, \quad \partial_t F_t(-2) \Big|_{t=0} = -\frac{1}{6}. \quad (\text{ZTC21})$$

For completeness, the heat equation and chain rule give $\partial_h f = (Af)''/(4A)$. At $h = 0$, $A\zeta$ is holomorphic at -2 whereas $1/A$ has a simple zero there, giving the first value. HTB5 gives the second as $2\zeta(-1) = -1/6$. The discrepancy holds on the actual initial arithmetic function. CFD21 further proves that all the zeros at $-2m$ remain simple along the real heat path, while its residue at 1 varies strictly; the actual Hurwitz path has constant residue 1 and values $-B_{2m+1}(1+t)/(2m+1)$ at those points.

Thus the valid original- ζ receiver consists of the original functions, their exact transform and inverse, the retained boundary component, the embedded local comparison modules, and all support labels. The scalar theta orbit by itself is a smaller observable. These proofs neither exhibit an off-line zero of the original ζ nor rule one out. They establish precisely which input and which deformation a subsequent estimate is estimating.

2 The actual adelic source and its retained prime boundary

The adelic restriction operation and its original measures are those of Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Section III (6)–(19), and Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, as specified in SWQ2–14. The homological complex used below is the standard Koszul complex, named for Jean-Louis Koszul. The standard regular-sequence mechanism is recorded by *The Stacks Project Authors*, Tag 062F, with original author TeX, lines 7264–7298, commit a04446e57ec1fbc252a871afcec7752fb2807b14. The contribution here is the exact application, connecting representatives, and receiving maps for these specified programme modules; the necessary Koszul exactness is proved below as well. All tensor products and quotients below are algebraic unless a different meaning is explicitly stated.

The precise public programme inputs are SWQ2–14, MRT1–20, and ER27–30. These pinned proof targets were checked against the local source versions.

2.1 The finite adelic basis and the exact original source

Put

$$G = \mathbb{Q}_{>0}^\times, \quad R = \mathbb{C}[G], \quad I = \ker(\varepsilon : R \longrightarrow \mathbb{C}), \quad H = \mathcal{S}_{\text{even}}(\mathbb{R}), \quad K = \widehat{\mathbb{Z}}^\times, \quad c_a = 1_{a\widehat{\mathbb{Z}}}. \quad (\text{ABR1})$$

The basis element $t_a \in R$ satisfies $t_a t_b = t_{ab}$ and $\varepsilon(t_a) = 1$. The finite additive measure has $\text{vol}(\mathbb{Z}_p) = 1$, and the real additive measure is Lebesgue measure. Thus

$$\text{vol}(a\widehat{\mathbb{Z}}) = \prod_p p^{-v_p(a)} = a^{-1} \quad (a > 0 \text{ rational}). \quad (\text{ABR2})$$

This equality uses the original positive real value of a and its prime factorization; the finite and real factors will both be retained.

The family $(c_a)_{a \in G}$ is a vector-space basis of $\mathcal{S}(\mathbb{A}_f)^K$. To prove spanning, first observe that a compactly supported locally constant function on $\mathbb{Q}_p$, invariant under $\mathbb{Z}_p^\times$, is constant on valuation shells. Choose integers $A \leq B$ such that its support is in $p^A \mathbb{Z}_p$ and it is constant on cosets of $p^B \mathbb{Z}_p$. If its values on the shells $A \leq n < B$ and on the last ball $p^B \mathbb{Z}_p$ are $d_A, \dots, d_B$, the function is exactly

$$d_A 1_{p^A \mathbb{Z}_p} + \sum_{n=A+1}^B (d_n - d_{n-1}) 1_{p^n \mathbb{Z}_p}.$$

The original Bruhat space on $\mathbb{A}_f$ consists of finite sums of restricted tensor products with $1_{\mathbb{Z}_p}$ outside a finite set of primes. Compact-unit averaging of a finite-place test is a finite linear operation: its support is contained in finitely many bounded local balls and its local-constancy subgroups have finite index there. Averaging therefore reduces to a finite quotient. Apply the preceding local formula at the finitely many relevant primes and expand the finite tensor products. Their balls are exactly c_a . For an invariant test averaging leaves the test itself fixed, which proves spanning.

For independence, take distinct $a_1, \dots, a_m \in G$ and partially order them by $a_i \preceq a_j$ when a_j/a_i is an integer. This is a partial order because both positive ratios can be integers only when they are one. Order this finite set by a linear extension. Evaluate its characteristic functions at the finite rational ideles $x_f = a_j$. Then

$$c_{a_i}(a_j) = 1 \iff a_i \preceq a_j.$$

Indeed $\mathbb{Q} \cap \widehat{\mathbb{Z}} = \mathbb{Z}$. The resulting matrix is lower triangular with diagonal one. Its determinant is one, proving independence. This proof also works for coefficients in any complex vector space, because back substitution uses only this finite triangular matrix.

Define

$$\Theta : R \otimes H \longrightarrow \mathcal{S}(\mathbb{A}_{\mathbb{Q}})^{K, \text{even}}, \quad \Theta(t_a \otimes h)(x_\infty, x_f) = h(x_\infty/a) c_a(x_f). \quad (\text{ABR3})$$

Here evenness refers to the original real variable x_∞ . Every target is a finite sum $\sum_a g_a(x_\infty) c_a(x_f)$ with $g_a \in H$. Set $h_a(x) = g_a(ax)$ to obtain its preimage in ABR3. Independence follows from the vector-valued independence of the c_a and the injectivity of each real dilation. Thus Θ is an isomorphism of vector spaces, with the displayed exact inverse. No change in the original real coordinate has been made in the target.

For $r \in G$ define full rational adelic dilation by $\alpha_r \xi(x) = \xi(r^{-1}x)$. The finite argument gives $c_a(r^{-1}x_f) = c_{ra}(x_f)$, while the real argument is $h(x_\infty/(ra))$. Consequently

$$\alpha_r \Theta(t_a \otimes h) = \Theta(t_{ra} \otimes h). \quad (\text{ABR4})$$

This is the left R -action. It is full diagonal rational dilation; it is not dilation of the real idele-class coordinate alone.

Let $m(h) = (h(0), \int_{\mathbb{R}} h(x) dx)$. Evaluation and integration of ABR3 give the two exact moments

$$\Theta(t_a \otimes h)(0) = h(0), \quad \int_{\mathbb{A}} \Theta(t_a \otimes h) = a \left(\int_{\mathbb{R}} h \right) \prod_p p^{-v_p(a)} = \int_{\mathbb{R}} h. \quad (\text{ABR5})$$

Thus $(\varepsilon_0, \varepsilon_1)\Theta = \varepsilon \otimes m$. In particular

$$M_0 = \ker(\varepsilon \otimes m), \quad H_{00} = \ker m, \quad \Theta(M_0) = \mathcal{S}(\mathbb{A})_0^{K, \text{even}}. \quad (\text{ABR6})$$

This is equality of the actual spaces with their two original endpoint conditions, not a comparison only after completion.

2.2 The complete algebraic coinvariant and its boundary

The two even Schwartz functions

$$h_0(x) = (1 - 2\pi x^2)e^{-\pi x^2}, \quad h_1(x) = 2\pi x^2 e^{-\pi x^2}$$

have moments $(1, 0)$ and $(0, 1)$. Indeed their evaluations are immediate and the real Gaussian integrals are $\int e^{-\pi x^2} = 1$ and $\int x^2 e^{-\pi x^2} = 1/(2\pi)$. The section $\sigma(u_0, u_1) = u_0 h_0 + u_1 h_1$ therefore splits m . It gives

$$M_0 = (R \otimes H_{00}) \oplus (I \otimes \sigma(\mathbb{C}^2)), \quad Q_0 = M_0/IM_0 \cong H_{00} \oplus (I/I^2) \otimes \mathbb{C}^2. \quad (\text{ABR7})$$

Indeed multiplication by I has exactly the two components $I \otimes H_{00}$ and $I^2 \otimes \sigma(\mathbb{C}^2)$.

For clarity the resulting boundary and inverse can be written without leaving a dependence on the chosen moment section. Put

$$W = \bigoplus_{p \text{ prime}} \mathbb{C} \ell_p, \quad \mathcal{B}_{\text{pr}} = W \otimes \mathbb{C}^2. \quad (\text{ABR8})$$

The map $[t_a - 1] \mapsto \sum_p v_p(a) \ell_p$ is an isomorphism $I/I^2 \rightarrow W$. For its proof retain the identity $t_{ab} - 1 = (t_a - 1) + (t_b - 1) + (t_a - 1)(t_b - 1)$. Modulo I^2 it gives additivity, and inversion gives the negative of a class. Prime factorization expresses each class as the stated sum. For independence the functionals $D_p(\sum_a c_a t_a) = \sum_a c_a v_p(a)$ satisfy $D_p(rs) = \varepsilon(r) D_p(s) + \varepsilon(s) D_p(r)$, and hence vanish on I^2 . They take value one on $[t_p - 1]$ and zero on the other displayed prime generators.

The canonical coordinates of $c = [\sum_a t_a \otimes h_a] \in Q_0$ are

$$\Phi(c) = \sum_a h_a \in H_{00}, \quad \beta(c) = \sum_p \ell_p \otimes \sum_a v_p(a) m(h_a). \quad (\text{ABR9})$$

Both maps kill IM_0 . For β , apply the preceding derivation identity to a product $r \sum_a t_a \otimes h_a$, $r \in I$; its value is $\sum_p D_p(r) \ell_p \otimes \sum_a m(h_a) = 0$ by the defining moment condition. The inverse of ABR9 sends

$$(h, \sum_p \ell_p \otimes u_p) \mapsto [t_1 \otimes h] + \sum_p [(t_p - 1) \otimes \sigma(u_p)]. \quad (\text{ABR10})$$

Only finitely many u_p are nonzero. Changing σ changes the latter expression by an element of $I \otimes H_{00} \subset IM_0$. To check the inverse on an arbitrary representative, subtract $t_1 \otimes \sum_a h_a$ and write the remainder as $\sum_a (t_a - 1) \otimes h_a$. Replacing each h_a there by $\sigma m(h_a)$ changes its class by IM_0 , and ABR8 then gives exactly ABR10. Thus the short exact sequence

$$0 \longrightarrow \mathcal{B}_{\text{pr}} \xrightarrow{j} Q_0 \xrightarrow{\Phi} H_{00} \longrightarrow 0 \quad (\text{ABR11})$$

is canonically split by $h \mapsto [t_1 \otimes h]$, and j is the section-independent second term of ABR10.

2.3 Actual periodization, injectivity, and exact inverse

For the original restriction map $E\xi(u, y) = \sum_{q \in \mathbb{Q}^\times} \xi(qy, qu)$, $u \in K$, $y > 0$, the finite condition in ABR3 is $q/a \in \widehat{\mathbb{Z}} \cap \mathbb{Q} = \mathbb{Z}$. Hence $q = an$ for $n \in \mathbb{Z} \setminus \{0\}$. It follows that

$$E\Theta(t_a \otimes h)(u, y) = \sum_{n \in \mathbb{Z} \setminus \{0\}} h(ny) = 2 \sum_{n \geq 1} h(ny) =: \mathcal{P}h(y). \quad (\text{ABR12})$$

Evenness of h supplies exactly the last factor two. The real and finite scales a have both participated in the equality; no finite restriction has been dropped. All sums are locally absolutely convergent with every real derivative by Schwartz decay. For $m = \sum_a t_a \otimes h_a \in M_0$, ABR12 is $E\Theta(m) = \mathcal{P}(\sum_a h_a)$.

The Fourier convention $\widehat{h}(\xi) = \int_{\mathbb{R}} h(x) e^{-2\pi i x \xi} dx$ gives the exact Poisson identity with both endpoints

$$\mathcal{P}h(y) = y^{-1} \widehat{\mathcal{P}h}(1/y) + y^{-1} \int_{\mathbb{R}} h(x) dx - h(0). \quad (\text{ABR13})$$

For $h \in H_{00}$ both displayed endpoint terms are zero, and $\widehat{h}$ also belongs to H_{00} because $\widehat{h}(0) = \int h = 0$ and $\int \widehat{h} = h(0) = 0$. At infinity the sums of every Euler derivative of $\mathcal{P}h$ decay faster than every power: choose an arbitrarily large Schwartz power $A > 1$ and bound their sums by a constant times $y^{-A} \sum n^{-A}$. ABR13 proves the corresponding estimates at zero. Therefore $\mathcal{P}h$ belongs to the original radial strong Schwartz test space. All powers of $\log y$ are also controlled by these bounds, so its Mellin transform is entire.

For $\Re s > 1$, absolute Fubini gives the original identity

$$\mathcal{M}(\mathcal{P}h)(s) = 2\zeta(s) \mathcal{M}_+ h(s), \quad \mathcal{M}_+ h(s) = \int_0^\infty h(x) x^{s-1} dx. \quad (\text{ABR14})$$

Indeed the absolute sum of integrals is $2\zeta(\Re s) \int_0^\infty |h(x)| x^{\Re s-1} dx < \infty$. The original Euler product proves that $\zeta(s) \neq 0$ in this half-plane: the logarithms of its factors converge absolutely because $\sum_p \sum_{k \geq 1} p^{-k\Re s}/k < \infty$. Thus division by the actual $2\zeta(s)$ is valid on the exact domain used below. There is no completed replacement of ζ .

For $\sigma > 1$, $g_\sigma(v) = e^{\sigma v} h(e^v)$ is Schwartz on $\mathbb{R}$. Every derivative is a finite sum of $e^{(\sigma+j)v} h^{(j)}(e^v)$. These decay exponentially at $-\infty$ and faster than any prescribed exponential at $+\infty$, using the original real Schwartz estimates. Consequently $\mathcal{M}_+ h(\sigma + it)$ decays faster than every power of t . Fourier inversion in v gives

$$h(x) = \frac{1}{4\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{\mathcal{M}(\mathcal{P}h)(s)}{\zeta(s)} x^{-s} ds, \quad x > 0, \quad \sigma > 1. \quad (\text{ABR15})$$

Every constant is explicit: Mellin inversion has $1/(2\pi i)$, and ABR14 contributes the additional $1/2$. To justify inversion directly, multiply the Fourier integrand by $e^{-\epsilon t^2}$. Its inverse is convolution with $(4\pi\epsilon)^{-1/2} e^{-v^2/(4\epsilon)}$, which tends to g_σ as $\epsilon \downarrow 0$. The proved rapid vertical decay permits dominated convergence on the Fourier side. This proves ABR15, including absolute convergence. Evenness recovers $h(x)$ for $x < 0$ and $h(0) = 0$ supplies the remaining point. In particular $\mathcal{P}$ is injective on H_{00} .

There is also an exact arithmetic inverse in the original variable:

$$h(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) \mathcal{P}h(nx), \quad x > 0. \quad (\text{ABR15a})$$

For fixed $x > 0$ and any $A > 1$, Schwartz decay bounds the absolute double sum after inserting ABR12 by a constant depending on x, A times $\sum_{n,m \geq 1} (nm)^{-A} = \zeta(A)^2$. It may therefore be regrouped by $k = nm$. Its coefficient of $h(kx)$ is $\sum_{n|k} \mu(n)$, which is one for $k = 1$ and zero for $k > 1$: prime factorization gives the finite product $\prod_{p|k} (1 - 1)$ in the latter case. This proves ABR15a with its factor $1/2$ and also proves convergence locally with every Euler derivative. It is a second exact inverse, not a truncation or a substitute for ABR14–15’s original zeta factor.

Let $V = ES(\mathbb{A})_0$ be the actual restriction image, and let V^K be its compact-unit invariant part. Then

$$V^K = ES(\mathbb{A})_0^{K, \text{even}} = \mathcal{P}(H_{00}). \quad (\text{ABR16})$$

For the nontrivial inclusion, take $F = E\xi \in V^K$. Average $\xi(x_\infty, vx_f)$ over $v \in K$. This is a finite average in the original Bruhat space, retains both moments, and has restriction equal to F . Next project to real-even functions by averaging x_∞ with $-x_\infty$. Under E , the real reflection equals the finite-unit translation $u \mapsto -u$, as seen by reindexing $q \mapsto -q$ in its absolutely convergent sum. Since F is K -invariant this projection still restricts to F . ABR6 now supplies a preimage in M_0 , whose sum of coefficients is in H_{00} . This proves ABR16 without replacing V by its closure or by a topological completion.

Full rational dilation is trivial after periodization, also directly by reindexing q/r in the rational sum. Therefore ABR12 induces a surjection on the algebraic coinvariant, with the exact kernel

$$0 \longrightarrow \mathcal{B}_{\text{pr}} \xrightarrow{j} Q_0 \xrightarrow{\rho = \mathcal{P}\Phi} V^K \longrightarrow 0. \quad (\text{ABR17})$$

Injectivity of $\mathcal{P}$ proves that this kernel is exactly the whole prime boundary, rather than merely containing it. A faithful joint receiver and its inverse are

$$\Omega(c) = (\rho(c), \beta(c)), \quad \Omega : Q_0 \xrightarrow{\sim} V^K \oplus \mathcal{B}_{\text{pr}}, \quad \Omega^{-1}(F, b) = [t_1 \otimes \mathcal{P}^{-1}F] + j(b). \quad (\text{ABR18})$$

The inverse $\mathcal{P}^{-1}$ is exactly ABR15.

Keep Connes’s original seminorm, without changing its measure, factor or weight:

$$\|c\|_\delta^2 = \int_0^\infty |y^{1/2} \rho(c)(y)|^2 (1 + \log^2 y)^{\delta/2} \frac{dy}{y}, \quad \delta > 1. \quad (\text{ABR19})$$

It is finite by the strong test estimates. Its weight is strictly positive. If the integral is zero, continuity gives $\rho(c) = 0$ everywhere, and ABR17 gives $c \in j(\mathcal{B}_{\text{pr}})$. The reverse inclusion follows from ABR17 as well. Thus ABR19 has precisely this algebraic nullspace. This identifies the lost coordinates but assigns no new norm to them and claims no equivalence with an unproved topological quotient.

2.4 Positive real ideles and the two endpoint characters

For $b > 0$, now with no rationality assumption, define $R_b h(x) = h(x/b)$ and let $\mathcal{R}_b = \text{id}_R \otimes R_b$ be the operator on $R \otimes H$. Its original moments obey

$$m(R_b h) = D_b m(h), \quad D_b = \begin{pmatrix} 1 & 0 \\ 0 & b \end{pmatrix}. \quad (\text{ABR19a})$$

Evaluation at zero is unchanged, and the real substitution $x = bu$ multiplies the additive integral by b . Thus $\mathcal{R}_b$ preserves M_0 , commutes with the full rational action, and descends to Q_0 . Under Θ it is precisely the real-idele action $\xi(x_\infty, x_f) \mapsto \xi(x_\infty/b, x_f)$; the finite support $a\widehat{\mathbb{Z}}$ has not changed.

In the exact coordinates ABR9, this action is

$$(\Phi, \beta) \mathcal{R}_b c = (R_b \Phi c, (\text{id}_W \otimes D_b) \beta c). \quad (\text{ABR19b})$$

This follows directly from the finite sums in ABR9. On the boundary inverse ABR10 it is also independent of the moment section: $R_b\sigma(u) - \sigma(D_b u) \in H_{00}$, so its product with $t_p - 1$ is in $I \otimes H_{00} \subset IM_0$. Consequently Ω is equivariant for real dilation on its V^K component and the two endpoint characters $1, b$ on its boundary component. These are the actual Connes endpoint characters $\mathbb{C} \oplus \mathbb{C}(1)$, with their displayed action; no purity theorem is inferred from this character calculation.

Both periodization conventions and their exact comparison are

$$\begin{aligned} E_{2007}(R_b h)(y) &= (\mathcal{P}h)(y/b), \\ E_{1998}(h)(y) &= y^{1/2} \mathcal{P}h(y), \\ E_{1998}(R_b h)(y) &= b^{1/2} E_{1998}(h)(y/b). \end{aligned} \tag{ABR19c}$$

The first equality follows term by term from ABR12. The last one retains $y^{1/2} = b^{1/2}(y/b)^{1/2}$ and therefore its exact factor $b^{1/2}$. Multiplication by the nowhere-zero function $y^{1/2}$ is explicitly inverted by multiplication by $y^{-1/2}$ on the original strong test spaces; it is not omitted in the seminorm. In particular that seminorm transforms exactly as

$$\|\mathcal{R}_{bc}\|_\delta^2 = b \int_0^\infty |E_{1998}(\Phi c)(y)|^2 \{1 + (\log y + \log b)^2\}^{\delta/2} \frac{dy}{y}. \tag{ABR19d}$$

This follows by the positive substitution in ABR19 and does not claim that the original weighted seminorm is dilation-invariant.

2.5 Every higher-degree connecting representative

Identify R with the Laurent polynomial algebra $\mathbb{C}[t_p^{\pm 1} : p \text{ prime}]$, whose polynomials involve only finitely many variables. Let $x_p = t_p - 1$ and use the ordered prime basis ℓ_p of W . The standard algebraic Koszul complex is

$$K_n(R) = R \otimes \Lambda^n W, \quad d(r \otimes \ell_{p_1} \wedge \cdots \wedge \ell_{p_n}) = \sum_{i=1}^n (-1)^{i-1} r x_{p_i} \otimes \ell_{p_1} \wedge \cdots \wedge \widehat{\ell_{p_i}} \cdots \wedge \ell_{p_n}. \tag{ABR20}$$

The indices here satisfy $p_1 < \cdots < p_n$. The two orders in which a pair of indices is removed have opposite signs and the coefficients commute, so $d^2 = 0$.

This is a free resolution of the augmentation module $\mathbb{C}$. For a finite set of primes the sequence of its x_p is regular: after any preceding variables have been set to one, the quotient is still a Laurent polynomial algebra in the remaining variables, and the next $t_p - 1$ is a nonzero divisor. The finite Koszul complex is exact in positive degrees by induction: adjoining one variable is the mapping cone of multiplication by that variable's x_p , whose kernel on the preceding degree-zero quotient is zero and whose cokernel is the further quotient. The full complex ABR20 is the filtered union of these finite complexes. A cycle uses finitely many wedge indices; it is a boundary already in the corresponding finite complex. Degree zero is $R/(x_p : p \text{ prime}) = \mathbb{C}$. This proves exactness also for the infinitely generated group, with only finite algebraic sums at every step.

Tensor this resolution with the coefficient modules in the exact sequence

$$0 \longrightarrow M_0 \longrightarrow R \otimes H \xrightarrow{\varepsilon \otimes m} \mathbb{C}^2 \longrightarrow 0.$$

Here $\mathbb{C}^2$ has trivial R -action. Tensoring each free Koszul degree preserves the short exact sequence. For the middle module, its Koszul homology is H in degree zero and zero in positive degrees, because the complex is $K_\bullet(R) \otimes H$. For the last module its differential is zero, and its degree- n homology is $\Lambda^n W \otimes \mathbb{C}^2$. Consequently

$$H_n(G, M_0) \cong \Lambda^{n+1} W \otimes \mathbb{C}^2 \quad (n \geq 1), \quad H_0(G, M_0) = Q_0, \tag{ABR21}$$

with ABR11 as the degree-zero exact sequence.

The exact connecting representative of an ordered element is

$$\begin{aligned} & \partial_n((\ell_{p_1} \wedge \cdots \wedge \ell_{p_{n+1}}) \otimes u) \\ &= \left[\sum_{i=1}^{n+1} (-1)^{i-1} (x_{p_i} \otimes \sigma(u)) \otimes \ell_{p_1} \wedge \cdots \widehat{\ell_{p_i}} \cdots \wedge \ell_{p_{n+1}} \right] \in H_n(G, M_0). \end{aligned} \quad (\text{ABR22})$$

Each coefficient is in M_0 because $\varepsilon(x_p) = 0$. This representative is the differential of $(t_1 \otimes \sigma(u)) \otimes \ell_{p_1} \wedge \cdots \wedge \ell_{p_{n+1}}$ in the middle module's complex, so it is a cycle. Changing a lift of u changes it by a boundary in the first complex. For $n \geq 1$ its map is an isomorphism: a cycle of the first complex becomes a boundary in the acyclic positive-degree middle complex, proving surjectivity; if a connecting class is a boundary, subtract its first-complex primitive from the chosen lift and use middle-complex acyclicity one degree higher, proving injectivity. Projection to the last complex has zero differential, so the latter argument forces the original exterior class to be zero. For $n = 0$ the representative is $[(t_p - 1) \otimes \sigma(u)]$, exactly j in ABR11. These proofs retain the original wedge order and sign.

The module V^K has trivial full-rational action, and its Koszul homology is $\Lambda^n W \otimes V^K$. Periodization kills every coefficient $x_p \otimes h$ in ABR22 separately, by ABR12. Thus the homology map induced by $E\Theta : M_0 \rightarrow V^K$ is zero in every positive degree. In degree zero it is exactly ABR17. The action $\mathcal{R}_b$ commutes with the differential ABR20. Under the connecting isomorphism ABR21 its action is precisely $\text{id}_{\Lambda^{n+1}W} \otimes D_b$. To verify the section-independence directly in ABR22, replace $R_b\sigma(u)$ by $\sigma(D_b u)$. Their difference is in H_{00} , and the resulting difference of cycles is the differential of $(t_1 \otimes (R_b\sigma(u) - \sigma(D_b u)))$ times the complete ordered wedge, which lies in the first complex itself. Thus every higher connecting class has the same two original endpoint characters, with its prime wedge unchanged.

2.6 The existing mass and endpoint sources under this map

The actual source spaces, moments, group action, quotient and coordinate maps in ABR1 and ABR6–10 agree with MRT1–5 and ER27–30, not just their dimensions. The MRT source heat kernel is

$$k_T(x) = (4\pi T)^{-1/2} e^{-x^2/(4T)}, \quad T > 0.$$

For its even finite coefficient source a , $A_T a(x) = \sum_n a_n k_T(x - n)$ is even Schwartz and has exactly

$$m(A_T a) = \left(\sum_n a_n k_T(n), \sum_n a_n \right) = (q_T(a), m(a)). \quad (\text{ABR23})$$

Therefore the actual class and antecedent are

$$\begin{aligned} B_{p,T}(a) &= [(t_p - 1) \otimes A_T a], \\ \Theta((t_p - 1) \otimes A_T a)(x) &= (A_T a)(x_\infty/p) c_p(x_f) - (A_T a)(x_\infty) c_1(x_f), \\ (\rho, \beta) B_{p,T}(a) &= (0, \ell_p \otimes (q_T(a), m(a))). \end{aligned} \quad (\text{ABR24})$$

Each such antecedent has both moments zero, including the separate finite Haar factor p^{-1} and the real integral factor p of its first summand. Its two periodizations are equal term by term after the exact arithmetic reindexing ABR12.

These finite Gaussian sources attain all of $\mathcal{B}_{\text{pr}}$. For a prescribed pair (u_0, u_1) and a fixed $T > 0$, retain $k_0 = k_T(0)$ and $k_1 = k_T(1)$ and take

$$a_0 = \frac{u_0 - k_1 u_1}{k_0 - k_1}, \quad a_1 = a_{-1} = \frac{k_0 u_1 - u_0}{2(k_0 - k_1)}, \quad a_n = 0 \quad (|n| > 1).$$

Here $k_0 > k_1$ and direct substitution gives ABR23 equal to (u_0, u_1) . Summing the finitely many selected prime classes therefore realizes every boundary vector, with the original kernel and coordinate conventions of MRT8.

For the Gaussian coefficient family $a_n^x = e^{-\pi n^2 x}$, $x > 0$, MRT13–14 proves that the same $A_T a^x$ is even Schwartz and gives

$$\beta B_{p,T}(a^x) = \ell_p \otimes \left((4\pi T)^{-1/2} \vartheta(x + 1/(4\pi T)), \vartheta(x) \right), \quad \rho B_{p,T}(a^x) = 0. \quad (\text{ABR25})$$

No heat time has been identified with the independent parameter x . The separately specified MRT supported-zero source operation $\widehat{E}$ still has its actual receiving matrix $J_T = \begin{pmatrix} 0 & k_T(0) \\ 0 & 1 \end{pmatrix}$ on these two moments, as proved in MRT11. Its periodization is zero before and after that operation. The present comparison does not redefine $\widehat{E}$ as the scalar supported-zero operation on a different common-label vector carrier.

For an arbitrary completed MRT input $(a, 0, m)$, MRT10 already defines its boundary through $(q_T(a), m)$. ABR10 supplies the Schwartz representative $\sigma(q_T(a), m)$ and hence an actual adelic antecedent for this boundary. This does not assert that $A_T a$ itself is Schwartz for every completed coefficient sequence.

For the ER source, keep its original real variable Z , time $t > 0$, and function

$$h_t(Z) = e^{-Z^2/(4t)} H_t(Z) \cos(Z/(2t)). \quad (\text{ABR26})$$

It is even Schwartz. The exact moments are

$$m(h_t) = \left(H_t(0), \frac{\sqrt{\pi t}}{8} e^{-1/(4t)} \right). \quad (\text{ABR27})$$

To retain the reason for the second constant, insert the original cosine integral

$$H_t(Z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(Zu) du.$$

The inner Gaussian integral is

$$\sqrt{\pi t} \{ e^{-t(u+1/(2t))^2} + e^{-t(u-1/(2t))^2} \} = 2\sqrt{\pi t} e^{-tu^2-1/(4t)} \cosh u.$$

Absolute Fubini follows from the Gaussian and the faster-than-Gaussian decay of the original Φ . The retained factor e^{tu^2} then gives $2\sqrt{\pi t} e^{-1/(4t)} H_0(i)$. ER3–5 gives $H_0(-i) = 1/16$ from the original limit

$$\frac{1}{16} \lim_{s \rightarrow 1} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s) = \frac{1}{16};$$

evenness gives $H_0(i) = H_0(-i)$. This proves ABR27 with the original zeta pole, Gamma factor, power of pi and factor 16 retained. No arithmetic Hurwitz-flow time is being substituted for t .

In ABR3 use this same real variable for x_∞ and h_t . The ER prime class has the actual antecedent and receiver

$$\begin{aligned} \Xi_{p,t}(x) &= h_t(x_\infty/p) c_p(x_f) - h_t(x_\infty) c_1(x_f), \\ \rho[(t_p - 1) \otimes h_t] &= 0, \\ \beta[(t_p - 1) \otimes h_t] &= \ell_p \otimes \left(H_t(0), \frac{\sqrt{\pi t}}{8} e^{-1/(4t)} \right) \neq 0. \end{aligned} \quad (\text{ABR28})$$

The last strict nonzero statement uses $t > 0$. Thus it is a nonzero element of the exact kernel ABR17, with both original source moments computed rather than assumed.

2.7 Common-label support and the full SWQ receiving quotient

For every complex vector space U keep

$$G_L(U) = \{(0, \lambda) : \lambda \in L\} \cup (U \times \{1_L\})$$

for the original arbitrary nontrivial bounded distributive lattice. Every linear map T lifts by $T^\uparrow(u, \lambda) = (Tu, \lambda)$, preserving addition with label join, scalar action with label meet, and compositions. The exact coinvariant quotient here is the same-label congruence

$$(m, \lambda) \sim (m', \lambda') \iff \lambda = \lambda' \text{ and } m - m' \in IM_0. \quad (\text{ABR29})$$

Its quotient is $G_L(Q_0)$. The further map $\rho^\uparrow$ has fibres with the same labels and amplitude differences in $j(\mathcal{B}_{\text{pr}})$. Therefore a top-labelled nonzero prime class from ABR24 or ABR28 maps to $(0, 1_L) = e$, not to $\tau = (0, 0_L)$. For a general zero-amplitude label its image is $(0, \lambda)$ with that same label. The joint map $\Omega^\uparrow$ is an isomorphism onto $G_L(V^K \oplus \mathcal{B}_{\text{pr}})$; the entire direct sum has one common label, not independently chosen component labels.

The Koszul amplitude complexes may be retained with these same common-label lifts. Their composites satisfy

$$d^\uparrow d^\uparrow(z, \lambda) = (0, \lambda) = e \cdot (z, \lambda). \quad (\text{ABR30})$$

Thus they are not ordinary complexes with a constant- τ zero map. The amplitude-zero cycles are exactly $G_L(\ker d)$, and the same-label congruence by $\text{im } d$ gives $G_L(H_n(G, M_0))$. Under the induced periodization the lifted classes ABR22 map to their retained zero labels in all positive degrees. This states the receiving maps without discarding the difference between amplitude zero and global absence.

Finally the range V^K in ABR17 is contained in the actual trivial-correspondence image V , not in a complementary spectral space. Let B_{CC} denote the ambient strong test algebra of SWQ and keep its original quotient $Q_{\text{CC}} = B_{\text{CC}}/\overline{V}$. The further composition

$$Q_0 \xrightarrow{\rho} V^K \longrightarrow B_{\text{CC}} \longrightarrow Q_{\text{CC}}$$

is the zero linear map because $V^K \subset V$. Its common-label lift retains each $(0, \lambda)$, in accordance with SWQ's same-label quotient; it does not identify e and τ . The exact faithful source receiver is ABR18, which retains the boundary rather than asserting an injection into this spectral quotient. No assertion that V is closed, no new metric on the prime boundary, and no positivity or Riemann-hypothesis theorem is made by this calculation.

3 The complete rational mixed boundary kernel

Original inputs and source provenance. The actual restriction map and its two zero endpoint conditions are those of Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, original author source `Yuri70v2.tex`, equations `TrMbf0`, `rangeCVdef`, `SCK` and Lemma `degmodifyV`. The full Gaussian-polynomial generator is retained below. HCB1–33 in `HURWITZ_CC_TEST_SPACE_BOUNDARY.tex` proves the original Hurwitz boundary coefficients with uniform differentiated remainders; those exact formulas, reproduced where used, are the analytic input. RMB1–10 in `CC_RATIONAL_MIXED_BOUNDARY.tex` constructs rational reflected antecedents and their full compact-unit character receiver. The present result classifies every finite mixed boundary with commensurable rational parameters; it does not replace the original functions, their scales, or the compact-unit coordinate.

The complete public predecessor proofs are HCB1–33 and RMB1–10, at the pinned edition whose expanded source matches the retained local reader.

3.1 All original parameters and the exact frequency map

Keep $b_0 > 0$ unchanged. Let $J \geq 1$, $a_j = 1 + t_j \in \mathbb{Q}_{>0}$, $r_j = b_j/b_0 \in \mathbb{Q}_{>0}$, and $\alpha_j \in \mathbb{C}$ for $1 \leq j \leq J$. Zero coefficients and repeated parameters are allowed. Define the original functions, without suppressing either summand, by

$$\begin{aligned} \eta(x) &= \pi x^2 (\pi x^2 - \tfrac{3}{2}) e^{-\pi x^2}, & g(x) &= 2\eta(x), \\ S(y) &= \sum_{j=1}^J \alpha_j \Psi_{t_j, b_j}(y) = \sum_{j=1}^J \alpha_j \sum_{n \geq 0} \left[2\pi^2 \left(\frac{(n + a_j)y}{b_j} \right)^4 - 3\pi \left(\frac{(n + a_j)y}{b_j} \right)^2 \right] e^{-\pi((n+a_j)y/b_j)^2}, & y > 0. \end{aligned} \tag{RMK1}$$

All sums converge absolutely, locally with every derivative, by Gaussian bounds; the finite sum in j introduces no limit interchange.

Choose a positive integer D divisible by the denominators of $1/r_j$ and a_j/r_j for every j . For example their least common multiple is an explicit permitted choice. Put

$$\begin{aligned} p_j &= Da_j/r_j \in \mathbb{Z}_{>0}, & q_j &= D/r_j \in \mathbb{Z}_{>0}, & B &= Db_0, \\ Q &= \text{lcm}(q_1, \dots, q_J), & K &= \max_{1 \leq j \leq J} \max(p_j - q_j, 0). \end{aligned} \tag{RMK2}$$

The original frequency comparison is the equality $(n + a_j)/b_j = (q_j n + p_j)/B$. Thus $n \mapsto q_j n + p_j$ is an exact bijection onto the positive progression starting at p_j . The full data $(\alpha_j, a_j, b_j, p_j, q_j)$ remain recorded; aggregating coincident frequencies does not identify their original presentations. Set, for every positive integer k ,

$$\begin{aligned} w_k &= \sum_{j=1}^J \alpha_j \mathbf{1}_{\{k \geq p_j, \ k \equiv p_j \pmod{q_j}\}}, \\ v_k &= \sum_{j=1}^J \alpha_j \mathbf{1}_{\{k \equiv p_j \pmod{q_j}\}}, \\ d_k &= - \sum_{j=1}^J \alpha_j \mathbf{1}_{\{1 \leq k < p_j, \ k \equiv p_j \pmod{q_j}\}}. \end{aligned} \tag{RMK3}$$

Extend v to every integer by the same formula. It is Q -periodic; $v_0 = v_Q$ is its residue-zero value, not its mean. Each omitted positive frequency in the j th progression is at most $p_j - q_j$. Consequently $d_k = 0$ for $k > K$, and

$$S(y) = \sum_{k \geq 1} v_k g(ky/B) + \sum_{k=1}^K d_k g(ky/B), \quad w_k = v_k + d_k. \tag{RMK4}$$

In particular no finite initial term was suppressed. If $p_j \leq q_j$ the corresponding initial defect is empty, including the case $p_j = q_j$. The frequency zero is absent from RMK1–4 because every $a_j > 0$; this does not impose $v_0 = 0$.

The exact ordinary generating function, for $|z| < 1$, and its rational continuation are

$$W(z) = \sum_{k \geq 1} w_k z^k = \sum_{j=1}^J \frac{\alpha_j z^{p_j}}{1 - z^{q_j}} = \frac{P(z)}{1 - z^Q} + D_+(z), \quad (RMK5)$$

$$P(z) = \sum_{r=1}^Q v_r z^r, \quad P^*(z) = z^Q P(1/z) = \sum_{r=1}^Q v_r z^{Q-r}, \quad D_+(z) = \sum_{k=1}^K d_k z^k.$$

These identities follow by summing convergent geometric series in each original progression and then the Q residue classes. They therefore give exact rational identities, including any cancellations caused by the original weights.

3.2 The full boundary generating function and its finite criterion

Use the Bernoulli polynomials with their original generating function

$$xe^{ax}/(e^x - 1) = \sum_{n \geq 0} B_n(a) x^n / n!.$$

HCB11–12 gives the actual differentiated asymptotic expansion of S with coefficients

$$C_m = \sum_{j=1}^J \alpha_j \frac{(-1)^{m+1} \pi^m B_{2m+1}(a_j)}{(m-1)! b_j^{2m}} = \frac{(-1)^{m+1} \pi^m}{(m-1)! B^{2m}} \beta_m, \quad m \geq 1,$$

$$\beta_m = \sum_{j=1}^J \alpha_j q_j^{2m} B_{2m+1}(p_j/q_j) \quad (RMK6)$$

$$= Q^{2m} \sum_{r=1}^Q v_r B_{2m+1}(r/Q) - (2m+1) \sum_{k=1}^K d_k k^{2m}.$$

The second expression for β_m follows either by substituting RMK4 and the Taylor series $g(x) = \sum_{m \geq 1} (-1)^m \pi^m (2m+1) x^{2m} / (m-1)!$, or by the Bernoulli shift identity in every progression. Every factor in the original C_m is retained in its exact comparison with β_m . For a finite sum, HCB's uniform remainder bounds may be summed; in particular

$$S \in \mathcal{B} \iff C_m = 0 \text{ for every } m \geq 1, \quad \mathcal{B} = \bigcap_{\gamma \in \mathbb{R}} y^\gamma \mathcal{S}(\mathbb{R}_+^*). \quad (RMK7)$$

For clarity, the reverse implication here is analytic: the zero coefficients and the differentiated remainder of arbitrary order give $|(y \partial_y)^r S(y)| = O(y^N)$ at zero for every r, N . The Gaussian bounds give all powers of decay at infinity. These are precisely the strong-space seminorms, including every derivative. The forward implication follows by successively taking the actual limits defining the asymptotic coefficients.

The full boundary generating function, including its constant, is

$$\mathcal{A}(u) = W(e^{u/B}) + W(e^{-u/B})$$

$$= \sum_{j=1}^J \alpha_j \left[\frac{e^{a_j u / b_j}}{1 - e^{u/b_j}} + \frac{e^{-a_j u / b_j}}{1 - e^{-u/b_j}} \right] \quad (RMK8)$$

$$= C + \sum_{m=1}^{\infty} \frac{2(-1)^m (m-1)!}{\pi^m (2m+1)!} C_m u^{2m}, \quad C = \sum_{j=1}^J \alpha_j (1 - 2a_j).$$

Each pair cancels its opposite simple principal parts at $u = 0$; the resulting function is holomorphic there. Substitution into the Bernoulli generating function proves the Taylor expansion for $|u| < 2\pi \min_j b_j$. Thus every positive even coefficient is specified, rather than merely a selected finite subset. As a function of $z = e^{u/B}$, the same expression is

$$W(z) + W(1/z) = \frac{P(z) - P^*(z)}{1 - z^Q} + D_+(z) + D_+(1/z). \quad (\text{RMK9})$$

The complete mixed boundary criterion is

$$\boxed{C_m = 0 \ (m \geq 1) \iff d_k = 0 \ (1 \leq k \leq K), \quad v_r = v_{Q-r} \ (1 \leq r < Q).} \quad (\text{RMK10})$$

To prove necessity, RMK8 makes the left side equivalent to $\mathcal{A}(u) = C$ near zero. The rational function in RMK9 therefore equals C identically: after clearing denominators, a polynomial vanishes on an open set and hence has all coefficients zero. At $z = 0$ the first term on the right of RMK9 is regular, as is $D_+(z)$. Its only possible negative powers are exactly $D_+(1/z) = \sum_{k=1}^K d_k z^{-k}$. Hence every d_k is zero. The remaining polynomial identity is $P - P^* = C(1 - z^Q)$. Its constant coefficient is $-v_Q$, so $C = -v_Q = -v_0$; the coefficients of z^r , $1 \leq r < Q$, are $v_r - v_{Q-r}$ and must be zero. Conversely these conditions give $P - P^* = -v_0(1 - z^Q)$ and $D_+ = 0$. RMK9 is then constant, so RMK8 proves every $C_m = 0$. This includes all complex cancellations between different original shifts and different original scales.

Here is a finite list of exact linear equations on the original weights, requiring no test of an infinite sequence:

$$\begin{aligned} \sum_{j=1}^J \alpha_j \mathbf{1}_{\{k < p_j, k \equiv p_j \ (q_j)\}} &= 0, \quad 1 \leq k \leq K, \\ \sum_{j=1}^J \alpha_j \left[\mathbf{1}_{\{r \equiv p_j \ (q_j)\}} - \mathbf{1}_{\{-r \equiv p_j \ (q_j)\}} \right] &= 0, \quad 1 \leq r \leq \lfloor (Q-1)/2 \rfloor. \end{aligned} \quad (\text{RMK11})$$

The first family is exactly $d_k = 0$ and the second is the nonrepeated part of the periodic reflection equations. The classes 0, and $Q/2$ when Q is even, are fixed by reflection and impose no amplitude equation. Their weights remain part of P and of the original function; they have not been set to zero. RMK11 is a finite matrix with integer entries acting on the original α_j .

There is also a sharp finite bound on the number of initial boundary coefficients that suffice. Set

$$M = K + \lfloor (Q-1)/2 \rfloor. \quad C_1 = \dots = C_M = 0 \iff C_m = 0 \text{ for every } m \geq 1. \quad (\text{RMK12})$$

When $M = 0$ the left side is the empty condition. To prove this, form the actual polynomial

$$N(z) = z^K [P(z) - P^*(z) + (1 - z^Q)\{D_+(z) + D_+(1/z) - C\}]. \quad (\text{RMK13})$$

All its powers are nonnegative and its degree is at most $2K + Q$. The constant in RMK8 is the actual value at $z = 1$. If the first M positive even coefficients vanish, its even holomorphic germ has $\mathcal{A}(u) - C = O(u^{2M+2})$. Since $u = B \log z$ has a simple zero at $z = 1$, multiplying by $z^K(1 - z^Q)$ shows that N vanishes at $z = 1$ to order at least $2M + 3$. This exceeds $2K + Q$ for both parities of Q . A polynomial of degree at most $2K + Q$ cannot have such a zero unless it is zero. RMK9–10 now give the result. For $Q = 1$ or $Q = 2$ and $K = 0$, all periodic weights are already even, and the empty condition correctly certifies the full kernel. For any original family, either RMK11 or the rational matrix $q_j^{2m} B_{2m+1}(p_j/q_j)$, $1 \leq m \leq M$, therefore computes the complete kernel by finite exact linear algebra.

The bound has the following precise dimension meaning. On the space of arbitrary Q -periodic amplitudes v and defects $d_1, \dots, d_K$, the obstruction coordinates are

$$\delta_r = v_r - v_{Q-r} \quad (1 \leq r \leq h), \quad h = \lfloor (Q-1)/2 \rfloor, \quad (d_1, \dots, d_K), \quad (\text{RMK14})$$

and the first $M = h + K$ comparison coefficients are

$$\beta_m = Q^{2m} \sum_{r=1}^h \delta_r B_{2m+1}(r/Q) - (2m+1) \sum_{k=1}^K d_k k^{2m}, \quad 1 \leq m \leq M. \quad (\text{RMK15})$$

This square rational matrix is invertible. Indeed RMK12 makes a zero output force a zero full boundary, and RMK10 then makes every δ_r, d_k zero; an injective linear map between finite spaces of equal dimension is invertible by elimination. Every such input occurs in an actual finite rational mixture: its periodic part is $\sum_{r=1}^Q v_r \Psi_{r/Q-1, B/Q}$, and $g(ky/B) = \Psi_{k-1, B} - \Psi_{k, B}$ represents each finite defect. Thus the dimension of this obstruction space is exactly M . Fewer than M scalar linear boundary tests cannot be injective on the whole space, by elementary rank. Symmetric periodic amplitudes are retained in (v, d) as kernel coordinates; RMK14 is not a replacement for that full original object.

For an explicit nonzero finite defect concealed by the first test, take $Q = 3$, $v_1 = 1$, $v_2 = v_3 = 0$, and $d_1 = 1/9$. Then

$$S(y) = \Psi_{-2/3, B/3}(y) + \frac{1}{9}[\Psi_{0, B}(y) - \Psi_{1, B}(y)], \quad \beta_1 = \frac{1}{3} - 3 \cdot \frac{1}{9} = 0, \quad \beta_2 = -\frac{5}{3} - 5 \cdot \frac{1}{9} = -\frac{20}{9}. \quad (\text{RMK16})$$

Here $K = 1$ and $M = 2$. Its second original boundary coefficient is $C_2 = 20\pi^2/(9B^4) \neq 0$. This calculation exhibits the finite initial term explicitly and shows why a first-coefficient test alone does not classify mixed families.

3.3 The exact actual adelic receiver and its endpoints

Suppose the equivalent conditions RMK10 hold. Define the finite locally constant function $\phi_v(x_f) = \mathbf{1}_{\widehat{\mathbb{Z}}}(x_f)v(x_f \bmod Q)$ on $\mathbb{A}_{\mathbb{Q}, f}$ and the actual adelic test

$$\xi_v(x_f, x_\infty) = \phi_v(x_f)\eta(x_\infty/B), \quad F_v(u, y) = \text{Tr } \rho(\xi_v)(u, y). \quad (\text{RMK17})$$

Every residue class modulo Q in $\widehat{\mathbb{Z}}$ has additive Haar mass $1/Q$. HCB3 computes $\eta(0) = 0$ and $\int_{\mathbb{R}} \eta = 3/4 - 3/4 = 0$. Therefore, retaining the two possibly nonzero finite weights,

$$\xi_v(0) = v_0\eta(0) = 0, \quad \int_{\mathbb{A}_{\mathbb{Q}}} \xi_v = \left(\frac{1}{Q} \sum_{r=1}^Q v_r \right) B \int_{\mathbb{R}} \eta = 0. \quad (\text{RMK18})$$

Thus $\xi_v \in \mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0$ in the original CCM source, and F_v lies in its actual restriction image $\mathcal{V} \subset \mathbf{S}(C_{\mathbb{Q}})$. Substitution into that restriction map gives the complete receiving formula

$$F_v(u, y) = \sum_{k \in \mathbb{Z} \setminus \{0\}} v(ku \bmod Q) \eta(ky/B) = \sum_{k \geq 1} v(ku \bmod Q) g(ky/B), \quad F_v(1, y) = S(y). \quad (\text{RMK19})$$

Only integer rationals contribute, since αu is integral at every finite place if and only if $\alpha \in \mathbb{Z}$. The second equality uses the proved evenness $v(-k) = v(k)$ and the evenness of η . Both signs of k supply exactly the factor $g = 2\eta$. The last equality uses $d = 0$ as well. This proves the existence of the actual finite adelic antecedent for every classified kernel function, with u retained, rather than merely a matching Mellin symbol.

For direct comparison with RMB1–3, take the pair with $a = r/q$, $1 - a = (q - r)/q$ and scale b , and choose $B = qb$. Its even weights are $v_k = \mathbf{1}_{k \equiv r \pmod{q}} + \mathbf{1}_{k \equiv -r \pmod{q}}$. If $\xi_{r, q, b} = 2\eta(x_\infty/b) \mathbf{1}_{r/q + \widehat{\mathbb{Z}}}(x_f)$ is RMB's antecedent, then

$$\xi_v(x) = \frac{1}{2} [\xi_{r, q, b}(x/q) + \xi_{r, q, b}(-x/q)], \quad \text{Tr } \rho(\xi_v) = \text{Tr } \rho(\xi_{r, q, b}). \quad (\text{RMK20})$$

The first equality follows directly from the two finite cosets and the unchanged real scale qb . For the second, multiplying the summation variable in $\mathbb{Q}^*$ by $1/q$ or $-1/q$ is a bijection in the absolutely

convergent restriction sum. Hence this comparison holds on the full (u, y) space, before identity slicing. It includes the order-two case by retaining the two identical coset contributions and their factor $1/2$.

All Mellin factors and both endpoint values can also be retained. For $\text{Re } s > 1$, set

$$\begin{aligned} I_g(s) &= \pi^{-s/2} [\Gamma(s/2 + 2) - \tfrac{3}{2}\Gamma(s/2 + 1)], \\ \mathcal{M}S(s) &= I_g(s) \sum_{j=1}^J \alpha_j b_j^s \zeta(s, a_j) \\ &= B^s I_g(s) \left[Q^{-s} \sum_{r=1}^Q v_r \zeta(s, r/Q) + \sum_{k=1}^K d_k k^{-s} \right]. \end{aligned} \tag{RMK21}$$

Absolute integration proves the first equality from RMK1 and the second from RMK4. HCB's continuation retains at $s = -2m$ exactly the residue C_m , including the factors in RMK6. When RMK10 holds, $S \in \mathcal{B}$, so its Mellin integral is entire; RMK21 is its exact meromorphic expression with every cancellation specified, not a newly chosen completed replacement. The finite values at $s = 0, 1$ are, respectively,

$$\begin{aligned} \int_0^\infty S(y) \frac{dy}{y} &= \sum_j \alpha_j (a_j/2 - 1/4) = -C/4 = v_0/4, \\ \int_0^\infty S(y) dy &= \tfrac{1}{4} \sum_j \alpha_j b_j = \frac{B}{4} \left(\frac{1}{Q} \sum_{r=1}^Q v_r \right). \end{aligned} \tag{RMK22}$$

The first two expressions in each line hold for the original mixture before assuming RMK10. For the last expression in the first line use $C = -v_0$ from its proof. For the second line note that the periodic mean is $\sum_j \alpha_j/q_j$ and $b_j = B/q_j$. Neither v_0 nor this mean must vanish.

For an additional direct expression of the cancellation mechanism, define the finite additive Fourier coefficients

$$\widehat{v}_\ell = \frac{1}{Q} \sum_{r=0}^{Q-1} v_r e^{-2\pi i \ell r/Q}, \quad v_r = \sum_{\ell=0}^{Q-1} \widehat{v}_\ell e^{2\pi i \ell r/Q}. \tag{RMK23}$$

Finite geometric-series summation proves the inverse. Evenness of v gives $\widehat{v}_{-\ell} = \widehat{v}_\ell$; this is an equality without conjugation and holds for complex weights. HCB24 proves $\widehat{g} = g$ with Fourier convention $e^{-2\pi i x \xi}$. Poisson summation for each of the finitely many additive characters therefore gives the exact formula

$$S(y) = \frac{B}{2y} \sum_{m \in \mathbb{Z}} \widehat{v}_m g\left(\frac{mB}{Qy}\right) = \frac{B}{y} \sum_{m \geq 1} \widehat{v}_m g\left(\frac{mB}{Qy}\right). \tag{RMK24}$$

To verify the factor, first write $S = \frac{1}{2} \sum_{k \in \mathbb{Z}} v_k g(ky/B)$; its zero summand is exactly $v_0 g(0) = 0$. For the character $e^{2\pi i \ell k/Q}$, Poisson gives $(B/y) \sum_n g((n - \ell/Q)B/y)$. Put $m = Qn - \ell$ and use $\widehat{v}_{-m} = \widehat{v}_m$. This proves both equalities in RMK24. The dual zero term is $\widehat{v}_0 g(0) = 0$, while $\widehat{v}_0 = Q^{-1} \sum_r v_r$ is retained as an independent endpoint quantity in RMK22. No zero residue or zero Fourier weight has been imposed by the classification.

3.4 The compact coordinate, exact convolution receiver, and supports

The full compact coordinate in RMK19 can be replaced invertibly by its finite character tuple, as in RMB4–9. Let $U_Q = (\mathbb{Z}/Q\mathbb{Z})^\times$, with its single element when $Q = 1$, and let χ range over all its characters. Give it Haar mass $1/\varphi(Q)$ per element, where $\varphi(1) = 1$. Define

$$\begin{aligned} F_\chi(y) &= \frac{1}{\varphi(Q)} \sum_{u \in U_Q} \overline{\chi(u)} F_v(u, y), \\ (P_\chi F_v)(u, y) &= \chi(u) F_\chi(y), \quad F_v(u, y) = \sum_\chi \chi(u) F_\chi(y). \end{aligned} \tag{RMK25}$$

Character orthogonality proves the inverse; it follows by multiplying the sum of a nontrivial character by a unit where its value differs from one, and by the usual finite character basis on the finite abelian group. Each projected term remains an actual restriction image: take the finite linear average of the adelic tests obtained from ξ_v by finite-unit multiplication. Both zero endpoint conditions persist under that average. No averaging of the original F_v to its principal component is performed.

For full level- Q strong functions $F(u, y) = \sum_{\chi} \chi(u) F_{\chi}(y)$ and $G(u, y) = \sum_{\chi} \chi(u) G_{\chi}(y)$, the original group convolution is

$$(F * G)(u, y) = \sum_{\chi} \chi(u) \int_0^{\infty} F_{\chi}(z) G_{\chi}(y/z) \frac{dz}{z}. \quad (\text{RMK26})$$

Insert the two finite expansions into its Haar integral. Orthogonality annuls terms with different characters, and the strong estimates make all real integrals absolutely convergent. The identity slice instead sums every character before convolution and thereby introduces mixed products. Thus RMK19 proves an identity slice and RMK25–26 prove the full invertible receiving map; these are not asserted to be the same morphism.

The radially of this particular full receiver also has a finite exact criterion:

$$F_v(u, y) \text{ is independent of } u \iff v_{ku} = v_k \text{ for all } k \in \mathbb{Z}, u \in U_Q. \quad (\text{RMK27})$$

For the only non-immediate direction, the function map $w \mapsto \sum_{k \geq 1} w_k g(ky/B)$ is injective on bounded sequences. If k_0 is the least index with $w_{k_0} \neq 0$, then

$$\lim_{h \rightarrow \infty} h^{-4} e^{\pi k_0^2 h^2} \sum_{k \geq 1} w_k g(kh) = 2\pi^2 k_0^4 w_{k_0} \neq 0. \quad (\text{RMK28})$$

For $h \geq 1$, the absolute tail is bounded by a constant times $\sum_{k > k_0} (k^4 + k^2) e^{-\pi(k^2 - k_0^2)h^2}$, dominated by its convergent value at $h = 1$ and tending termwise to zero. The single leading term has exactly the displayed limit. Apply this injectivity to $w_k = v_{ku} - v_k$ in RMK19 to prove RMK27.

The unit orbits on residues modulo Q are exactly the sets with fixed $\gcd(k, Q)$. To see transitivity, after factoring out that $\gcd d$ choose the prescribed unit ratio modulo Q/d and lift it to a unit modulo Q by the Chinese remainder theorem at each prime power. Thus RMK27 is equivalent to $v_k = A_{\gcd(k, Q)}$ for numbers A_d , $d \mid Q$, with $A_Q = v_0$. In that case put

$$c_d = \sum_{e \mid d} \mu(d/e) A_e, \quad v_k = \sum_{d \mid Q} c_d \mathbf{1}_{d \mid k}, \quad F_v(u, y) = S(y) = \sum_{d \mid Q} c_d f_{B/d}(y). \quad (\text{RMK29})$$

Here μ is the integer Möbius function. The divisor identity $\sum_{e \mid n} \mu(e) = 0$ for $n > 1$ and 1 for $n = 1$, proved by expanding $\prod_{p \mid n} (1 - 1)$, proves the inverse formula for v . Substitution into RMK19 gives the exact scales B/d and every coefficient in RMK29. In the absence of this unit invariance, the nonprincipal even components in RMK25 remain present. No nonexistence claim about some different radial antecedent is made.

Finally retain the full support data. For an arbitrary nontrivial bounded distributive lattice L , use $G_L(W) = \{(0, \lambda) : \lambda \in L\} \cup (W \times \{1_L\})$ with join in addition and meet in scalar multiplication, as in HCB30. A complex-linear map T has the exact common-label lift $G_L(T)(w, \lambda) = (Tw, \lambda)$; its zero-amplitude inverse image is $G_L(\ker T)$, not an assertion that its output is τ . Thus all finite linear maps RMK3, RMK6, RMK11 and RMK25, and the inverse RMK25, admit these lifts. The original data may also carry separate labels on each coefficient $z_j = (\alpha_j, \lambda_j) \in G_L(\mathbb{C})$. In that case retain the entire tuple $(z_j)_{j=1}^J$ and its exact incidence images:

$$\begin{aligned} \tilde{w}_k &= \sum_{j: k \geq p_j, k \equiv p_j(q_j)} z_j, & \tilde{v}_k &= \sum_{j: k \equiv p_j(q_j)} z_j, \\ \tilde{d}_k &= (-1, 1_L) \sum_{j: 1 \leq k < p_j, k \equiv p_j(q_j)} z_j, & \tilde{o}_r &= \tilde{v}_r + (-1, 1_L) \tilde{v}_{-r}. \end{aligned} \quad (\text{RMK30})$$

Empty sums are $\tau = (0, 0_L)$. These are finite sums at each frequency, so no infinite join or completeness of L is assumed. Their amplitudes are the coefficients and equations already proved; their labels are exactly the joins of the contributing original labels. In particular a repeated contribution in the difference $\tilde{o}_r$ remains a supported cancellation and is not deleted as a structural zero. The original labelled tuple is not identified with the generally non-injective collection of amplitude equations. The complete function receiver is

$$\sum_{j=1}^J (\alpha_j \Psi_{t_j, b_j}, \lambda_j) = (S, \Lambda), \quad \Lambda = \bigvee_{j=1}^J \lambda_j. \quad (\text{RMK31})$$

Only this finite join is needed. When evaluating a particular boundary coefficient, use the supported scalar $(\kappa_{m,j}, 1_L)$ with $\kappa_{m,j} = (-1)^{m+1} \pi^m B_{2m+1}(a_j) / ((m-1)! b_j^{2m})$, including its evaluated zero cases. The coefficient receiver is then (C_m, Λ) for every m . When RMK10 holds it is $(0, \Lambda)$; for any nonzero original coefficient α_j this is $e = (0, 1_L)$, not τ . The finite criterion tests amplitudes, preserves every original input label, and does not turn mixed cancellations into absence. The compact-character map and its inverse act on one common-labelled direct sum when that is the chosen input carrier; they do not assign independent labels to components of that common-labelled object.

The incidence construction also gives the exact support correction for RMK4 when the original coefficients have separate labels. Put

$$\begin{aligned} \mu_k &= \bigvee_{j: k \geq p_j, \ k \equiv p_j(q_j)} \lambda_j, & \sigma_k &= \bigvee_{j: 1 \leq k < p_j, \ k \equiv p_j(q_j)} \lambda_j, \\ \nu_k &= \bigvee_{j: k \equiv p_j(q_j)} \lambda_j = \mu_k \vee \sigma_k, & \tilde{v}_k + \tilde{d}_k &= \tilde{w}_k + (0, \sigma_k). \end{aligned} \quad (\text{RMK32})$$

Each join is finite, and the join of an empty set is 0_L . The periodic incidences partition into their retained and omitted initial incidences, proving $\nu_k = \mu_k \vee \sigma_k$. Both sides of the last equality have amplitude w_k by RMK3; their labels are respectively $\nu_k \vee \sigma_k = \nu_k$ and $\mu_k \vee \sigma_k = \nu_k$. This proves the equality in the actual support carrier. The condition $d_k = 0$ in RMK10 is an amplitude condition and does not set $\sigma_k = 0_L$. Likewise equality of the reflected amplitudes does not erase the label on $\tilde{o}_r$. The retained tuples $(\tilde{w}_k, (0, \sigma_k))$ and the original labelled inputs specify the comparison without identifying a supported cancellation with absence. On the common-label function carrier, RMK4 lifts directly with its one unchanged label; no identification of these two support presentations is needed or asserted.